

much as 25,000–45,000 components per hour (CPH), with mount precision as good as 30 microns. High-speed changeover facilities, intelligent tape feeders and conveyors, 5-megapixel in-line inspection systems, high-speed and high-resolution automatic optical inspection (AOI) systems, soldering robots, automatic high-speed printers with 12 seconds cycle time and 10 micron repeatable accuracies at fine pitch, six channel profiler packs for reflows, etc, are some of the examples of latest technology available today.

Apart from the non-stop technological upgrades happening to SMT equipment, we can also see some significant trends in the field—increased parallelisation and flexibility, high levels of software integration, environment-friendly and green processes (RoHS-based production), handling of smaller components, small yet intelligent machines, etc.

Let us take a quick look at what is hot and what is not in SMT equipment.

Fast operation

One of the keys to achieving faster time-to-market is faster operation of placement machines (also known as pick-and-place machines or chipshooters). There was a time when the placement machine was considered as a bottleneck in the SMT workflow. However, machines have become much faster now. The current generation of placement machines operate at an average of 30,000 CPH. Some 4-gantry machines can handle even 120,000 CPH such as SIPLACE X4i machine.

Dedicated machines that simultaneously pick-and-place components on a board are considered the fastest. But in this time and age, one requires flexibility as well and speed alone will not suffice. Hence some machines today adopt an optimal balance of simultaneous and sequential operation (for example, simultaneous picking of components and sequential placement to enable flexibility and customisation). In addition, it is possible to combine dedicated and customisable pick-and-



Assembléon's A-series pick-and-place machine

place in a process in order to achieve the right mix of flexibility and speed.

Build-to-order

According to a Frost & Sullivan report published last year, the SMT equipment market has been significantly hit by recession. The good news is that the report also identifies certain steps that equipment manufacturers can take to ensure that they survive this tough period. One of the key factors is to induce as much flexibility as possible into their equipment—in volume as well as mix. The reason for this is that the electronics assembly market is expected to be dominated by EMS providers in the future, and flexibility is an important parameter affecting their buying decision.

"We at Rangsons choose the equipment based on flexibility and adaptability for different production requirements. This enables us to provide integrated solutions to our customers. For example, machines which enable us to place higher-end components onto very thin boards to flexible boards with required speed and accuracy," says Bhatnagar.

Jack Chua, vice president, Siemens Electronics Assembly Systems, explains this trend in more detail. "Last year's market developments and the exceptional demands of global markets were clear signals that the players in the electronics industry must change their

approach. They can no longer afford to focus on individual processes or single machine but must consider the SMT process in its entirety. Modern build-to-order concepts are the answer to this paradigm shift. The electronics manufacturing processes need much more flexibility in order to raise performance, reduce inventories and speed up new product introductions," he adds.

Scalability

One important subset of flexibility is scalability of SMT machines because demand is not constant. Most top players including Siemens and Assembléon provide extremely modular SMT equipment. Hence a manufacturer can initially buy only what is required for his day-to-day operations and then expand by buying or renting extra components.

Siemens' SIPLACE SX features capacity-on-demand, which the company claims is the world's first placement machine that lets the user scale flexibility and performance independent of each other. In addition, other new innovations like SIPLACE MultiStar CPP head, SIPLACE LES software solution and flexible services like 'rent-a-gantry' make SMT lines more scalable than ever before and open the door to a totally new set of capabilities for order-oriented production processes. Earlier this year, SIPLACE also introduced the flexible production concept with its SIPLACE Quad Lane, enabling the